

A Three-Plane Counterexample to Problem 5.4 of arXiv:1508.06687

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Problem and answer

Cahill, Casazza, Jasper, and Woodland ask the following converse on page 18, Problem 5.4, of *Phase Retrieval* [1]. If a family $\{W_i\}_{i=1}^M$ of subspaces of $\mathcal{H}_N$ still fails phase retrieval after adjoining *every* possible subspace W_{M+1} , must there be orthonormal bases of the W_i whose union has a partition F_1, F_2 with

$$\dim \text{span } F_1 \leq M - 2, \quad \dim \text{span } F_2 \leq M - 1?$$

The answer is no, already for three proper planes in dimension three. Since our example has $M = N = 3$, it also answers the version in which the two printed bounds are read as $N - 2, N - 1$.

Theorem 1. *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let e_1, e_2, e_3 be the standard basis of $\mathbb{F}^3$, and set*

$$W_1 = e_1^\perp, \quad W_2 = e_2^\perp, \quad W_3 = (e_1 + e_2 + e_3)^\perp.$$

Then:

- for every subspace $W_4 \subseteq \mathbb{F}^3$, the four subspaces W_1, W_2, W_3, W_4 fail phase retrieval;*
- for every choice of orthonormal bases B_i of W_i , no partition $B_1 \dot{\cup} B_2 \dot{\cup} B_3 = F_1 \dot{\cup} F_2$ satisfies*

$$\dim \text{span } F_1 \leq 1, \quad \dim \text{span } F_2 \leq 2.$$

Consequently these three planes are a counterexample to Problem 5.4.

Proof intuition

Lift a signal x to the rank-one matrix xx^* . Squared projection norms become linear measurements,

$$\|P_W x\|^2 = \text{tr}(P_W x x^*).$$

Four measurements cannot avoid the determinant-zero cone inside the six-dimensional real space of real symmetric 3×3 matrices: their kernel has dimension at least two, and an odd cubic vanishes on every real two-plane. Our first two planes rule out semidefinite kernel points, so the resulting singular point is an indefinite rank-two difference $xx^* - yy^*$.

The partition obstruction has a different geometry. If its two spans are a line L and a plane H , then each two-dimensional W_i either equals H or has an orthonormal basis split between L and H . The three displayed planes force one W_i to equal H , after which their three pairwise intersection lines violate the required orthogonality.

No fourth subspace can work

Let P_i be the orthogonal projection onto W_i , including an arbitrary W_4 . We work in the six-dimensional real vector space $\text{Sym}_3(\mathbb{R})$ even when $\mathbb{F} = \mathbb{C}$. Each Hermitian P_i defines a real linear functional there, and we set

$$\mathcal{A}(Q) = (\text{tr}(P_1Q), \dots, \text{tr}(P_4Q)).$$

Thus $\dim_{\mathbb{R}} \ker \mathcal{A} \geq 2$. Choose a real two-dimensional subspace $E \subseteq \ker \mathcal{A}$. On the unit circle of E , the continuous function $Q \mapsto \det Q$ is odd:

$$\det(-Q) = -\det Q.$$

The intermediate value theorem therefore gives a nonzero $Q \in E$ with $\det Q = 0$.

There is no nonzero positive semidefinite matrix in $\ker \mathcal{A}$. Indeed, if $Q \geq 0$ and $\text{tr}(P_iQ) = 0$, then

$$0 = \text{tr}(P_iQ) = \|P_iQ^{1/2}\|_{\text{HS}}^2, \quad \text{ran } Q \subseteq \ker P_i = W_i^\perp.$$

Using only $i = 1, 2$ gives

$$\text{ran } Q \subseteq W_1^\perp \cap W_2^\perp = \text{span}(e_1) \cap \text{span}(e_2) = \{0\}.$$

The same argument applied to $-Q$ excludes nonzero negative semidefinite matrices. Hence our nonzero singular Q is indefinite. It has rank two and a spectral decomposition

$$Q = xx^* - yy^*$$

for nonzero orthogonal real vectors x, y . Since $Q \in \ker \mathcal{A}$,

$$\|P_ix\|^2 - \|P_iy\|^2 = \text{tr}(P_iQ) = 0, \quad i = 1, 2, 3, 4.$$

But orthogonal nonzero x, y cannot differ by a real sign or a complex phase. Thus the four subspaces fail phase retrieval. This proof permits an arbitrary real or complex fourth subspace.

The requested partition cannot exist

Assume to the contrary that orthonormal bases B_i and a partition F_1, F_2 exist. Write

$$L = \text{span } F_1, \quad H = \text{span } F_2, \quad \dim L \leq 1, \quad \dim H \leq 2.$$

Each W_i is two-dimensional. Its two basis vectors cannot both lie in L . If both lie in H , then $W_i = H$. Otherwise exactly one lies in each part; in that case $L \subset W_i$, the line $H \cap W_i$ is generated by the F_2 basis vector, and

$$L \perp (H \cap W_i). \tag{1}$$

The three normals $e_1, e_2, e_1 + e_2 + e_3$ are linearly independent, hence $W_1 \cap W_2 \cap W_3 = \{0\}$. Therefore not all three bases can split between L and H ; at least one, say that of W_k , lies wholly in H , forcing $H = W_k$. The other two planes, say W_i, W_j , must split, and their common F_1 line is necessarily

$$L = W_i \cap W_j.$$

There are three possibilities for k , all impossible:

$$\begin{aligned} k = 1 : \quad & W_2 \cap W_3 = \text{span}(e_1 - e_3), & W_2 \cap W_1 = \text{span}(e_3), (e_1 - e_3)^* e_3 = -1; \\ k = 2 : \quad & W_1 \cap W_3 = \text{span}(e_2 - e_3), & W_1 \cap W_2 = \text{span}(e_3), (e_2 - e_3)^* e_3 = -1; \\ k = 3 : \quad & W_1 \cap W_2 = \text{span}(e_3), & W_1 \cap W_3 = \text{span}(e_2 - e_3), e_3^* (e_2 - e_3) = -1. \end{aligned}$$

In each row the last nonzero inner product contradicts (1) for the indicated plane. This proves the second claim and completes the proof of Theorem 1.

Verification and scope

The script `code/verify_counterexample.py` checks the three projection matrices and the three intersection-line obstructions exactly. It also tests 102 rationally generated fourth subspaces of all ranks and finds an indefinite singular lifted-kernel witness in every case. This computation is only a sanity check; the determinant and incidence arguments above prove the universal statements.

The result is a negative answer, not a classification of one-subspace completability. A bounded search for the exact Problem 5.4 and Corollary 5.3 wording found the source but no later resolution; novelty should still receive expert literature review.

References

- [1] J. Cahill, P. G. Casazza, J. Jasper, and L. M. Woodland, *Phase Retrieval*, arXiv:1508.06687 (2015), arXiv:1508.06687.